JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Artificial Intelligence
"A Place to Learn; A Chance to Grow"
2024-25 (Odd Sem)

VISION	MISSION
"To Produce Competent Professionals equipped with technical knowledge and commitment for satisfying the needs of society."	1. To impart advanced knowledge with an inclination towards Research with well-equipped Labs. 2. To develop an ability to work ethically and Responsive towards the need of society.

CLASS TEST-2 EXAMINATION
Course: B. Tech. In Computer Science & Engineering
Subject: Computer Architecture & Organization
Subject Code:CS3T003
Year/Semester:3rd Semester (2nd Year)
Max Marks: 20 **Date: 08/10/2024** **Duration:- 1 Hr.**

Instructions to the Students:
1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:
1. Describe the fundamental organisation of a computer system
2. Interpret the functional architecture of computing systems. (Understanding)
3. Explain addressing modes, instruction formats and program control statements.
4. Describe basic concept of parallel computing and Describe fundamentals concepts of pipeline and vector processing
5. Identify, compare and assess issues related to ISA, memory, control and I/O functions. (Applying, Analysing, Evaluating)

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define virtual memory.	L1/CO2	1
Q.2	Define communication device modem.	L1/CO1	1
Q.3	What is software and its types.	L2/CO1	1
Q.4	State 3 steps of control sequence of fetch cycle.	L1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Answer in one two three sentence.**

Q.5	Explain types of memory with example.	L1/CO1	2
Q.6	Perform addition of two positive number (+20, +30).	L1/CO1	2
Q.7	Convert $(11101)_2$ binary number to decimal number format.	L2/CO1	2
Q.8	Sketch flowchart of Booth;s algorithm.	L2/CO1	2

SECTION C

1. **Solve any two questions.**

Q.9	Explain Auto-increment and decrement addressing mode.	L2/CO1	4
Q.10	Describe memory management requirements.	L2/CO4	4
Q.11	Compare hardwired control unit and micro-programmed control unit.	L4/CO3	4